

Horn Antenna Mount Assembly Instructions

Introduction

Thank you for purchasing the Horn Antenna Mounting Kit. The parts contained in this kit are used for attaching the fiberglass gusset plates included with the Horn antenna to the Horn antenna trailer hitch mount. Please be sure to consult the parts list below and identify the different components. If you find anything missing contact your GSSI sales representative at (603) 893-1109, Monday-Friday 8:00 am to 5:00 pm, EST.

Time required for assembly: 15-30 minutes.

Tools required for assembly: 1 adjustable wrench and ratchet with 1 1/8" socket (not included).

Parts List

Your Horn Antenna Mounting Kit comes with the following parts (Figures 1 and 2):

- | | |
|------------------------------------|--|
| (2) ratchet straps | (16) M10 flat washers |
| 1 bag containing (10) shim washers | (2) M12X20 hex bolts |
| (2) 3/8" eyebolts | (2) M12 flat washers |
| (2) 3/8" poly hex locknuts | (2) M12 split washers |
| (4) 3/8" flat washers | (1) Horn antenna trailer hitch mount
(Figure 2) |
| (8) M10X40 hex bolts | |
| (8) M10 poly hex locknuts | |

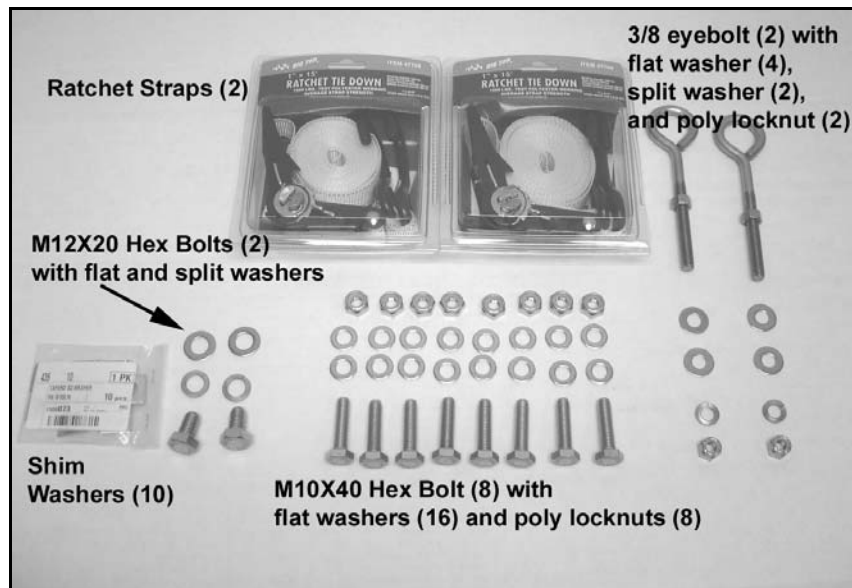


Figure 1: Mounting Hardware.

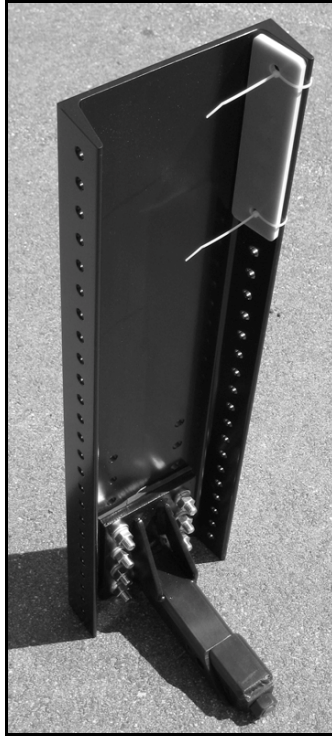


Figure 2: Horn Antenna Trailer Hitch Mount.

Assembly

Step 1: Attach Horn antenna trailer hitch mount to the vehicle. The hitch mount is designed to fit into a Class 3 trailer hitch. Cut the zip ties that hold the fiberglass rail stabilizer and set the rail stabilizer aside. Insert the draw bar into the hitch. You will notice holes on the sides of the split mount bar (Figure 3). Line up these holes with the two holes on the side of the hitch receiver. Tighten the 1 1/8" nut on the front of the mount. This will pull in the draw bar and secure the mount to the receiver.

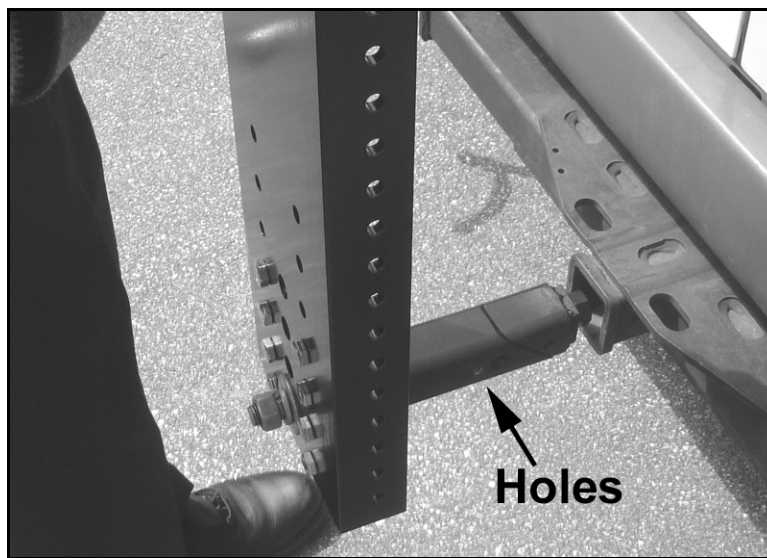


Figure 3: Insert mount into hitch receiver.

Step 2: Take the M12X40 bolt and put the flat washer on the bolt and then the split washer. Insert the bolt into the holes on the side of the hitch receiver and tighten securely (Figure 4). This is a safety precaution and will prevent the hitch mount from loosening and falling out of the receiver. Do the same on the other side of the receiver.

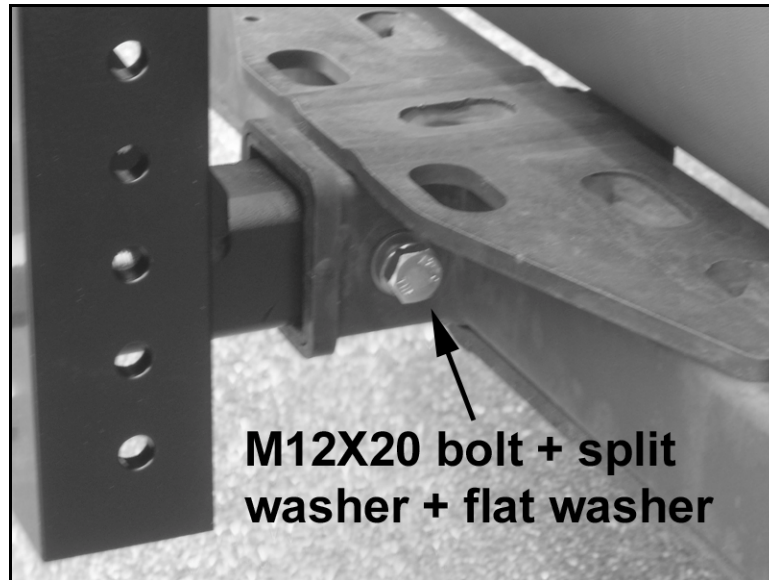


Figure 4: Safety bolt assembly.

Step 3: Attach the fiberglass rails to the horn antenna and the fiberglass gusset plates to the rails (Figure 5). The rails and the gussets were included with your horn antenna assembly. A detailed description of this procedure can be found in the GSSI Handbook for Road Structures.

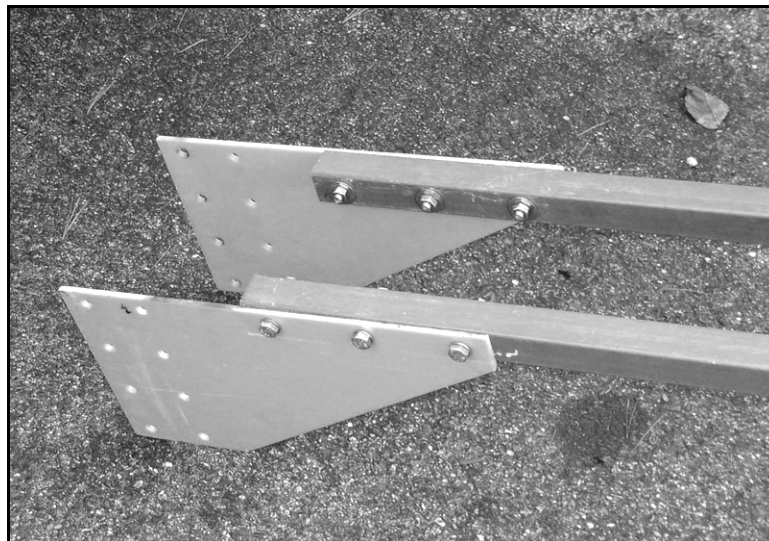


Figure 5: Gussets attached to rails

Step 4: Attach the fiberglass rail stabilizer to the rails with the 3/8" eyebolts. Secure with the 3/8" flat washers and 3/8" hex locknuts (Figure 6). The eyebolts are also the attachment point for the ratchet straps.

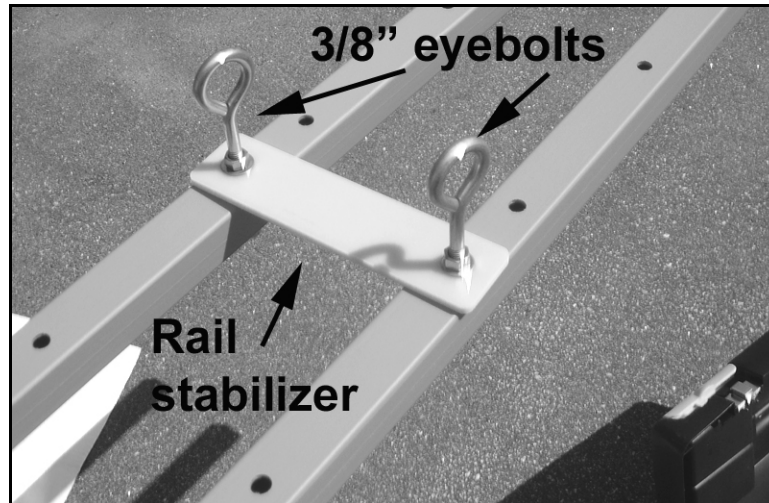


Figure 6: Rail stabilizer.

Step 5: Attach the antenna assembly to the hitch mount (Figure 7). The distance from the bottom of the antenna to the road surface should be approximately 18" (50 cm). Test fit the antenna assembly to the hitch mount to choose the correct hole matchup. Slide one M10 flat washer over one of the M10X40 hex bolts and then slide that bolt through the gusset plate and hitch mount (Figure 7).

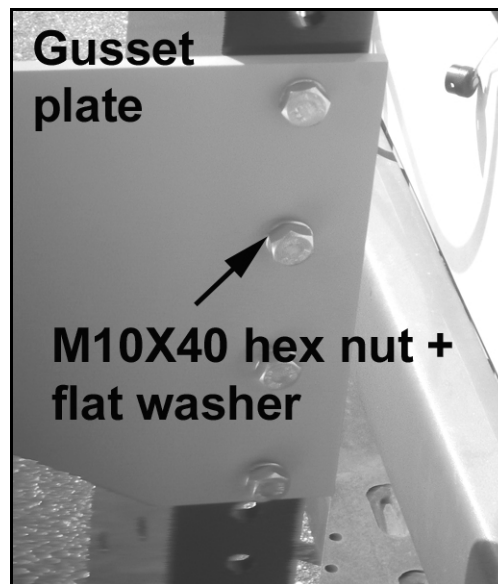


Figure 7: Attach antenna assembly.

Step 6: Attach antenna assembly to the antenna mount. Take one of the shim washers and slide it over the M10X40 bolt so that the angled side is against the mount wall (Figure 8). Note that the shim washer is designed to compensate for the angle in the hitch mount. Rotate the shim to compensate for the angle. Attach with a flat washer and M10 poly hex nut.



Figure 8: Place shim washer.

Step 7: Attach ratchet straps and tighten (Figure 9). These will help to stabilize the antenna. Pick a secure attachment point on the vehicle and when attaching the straps, position them so that the metal buckle is as far as possible from the antenna. This attachment point may experience considerable stress while surveying. Be careful to avoid over-tightening the ratchet straps. Over-tightening may cause permanent damage to the mounting rails.

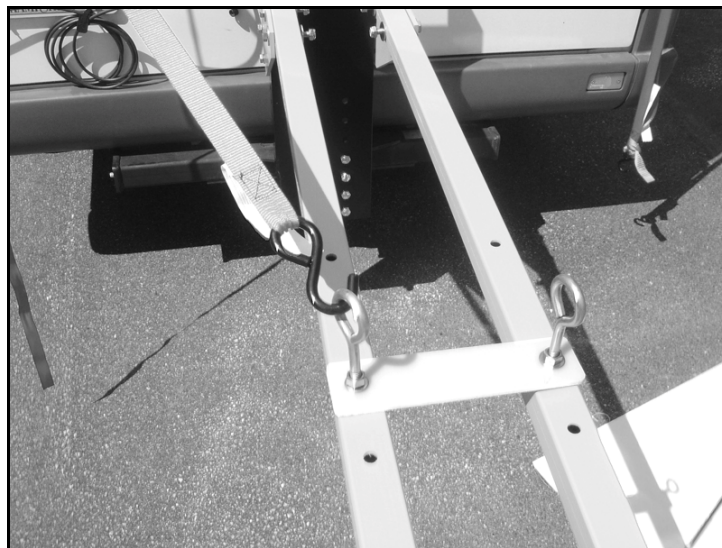


Figure 9: Attach ratchet straps.